

DIGIWIZARD – A Modular Multi-Dev Tablet

Priyadharshan P

*Computer Science & Engineering
Sathyabama Institute of Science and
Technology
Chennai, India
ORCID ID: 0009-0001-7020-9809
priyadharshansharma@gmail.com*

Monish Anand J

*Computer Science & Engineering
Sathyabama Institute of Science and
Technology
Chennai, India
ORCID ID: 0009-0005-0427-082X
monishanand.j@gmail.com*

Ms. T.Anandhi

*Computer Science & Engineering
Sathyabama Institute of Science and
Technology
Chennai, India
anandhi.cse@sathyabama.ac.in*

Abstract— The goal is to produce a tablet-like gadget on the base of a Raspberry Pi with functionalities such as Infrared and emulation of Radio Frequency Identification and Near-Field Communication. This tablet should be powered by a custom-built operating system with user-friendliness towards varied applications. The basic structure comprises a Raspberry Pi Model 4 or more advanced versions, a touchscreen, and modules that ensure Infrared and RFID/Near Field Communication. IR emulation This feature allows the gadget to work as a universal remote control, which replicates commands for different devices such as TVs and audio systems. On the other hand, the RFID/NFC module is employed to read, write, or emulate tags. These are useful in access applications and data transfers. Potential applications of this device may be a universal remote, smart home hub, educational tool, or prototype development platform for IoT projects. The project addresses problems in hardware compatibility, software development, power consumption, and security, making it a robust and efficient solution. This project aims to create a multifunctional device using Raspberry Pi capabilities with wide applicability, showing innovations both in electronics and wireless communication.

Keywords— Raspberry Pi, tablet, IR, RFID, NFC, custom OS, universal remote, smart home, IoT, hardware compatibility

I. INTRODUCTION

Over the last few years, technology convergence has actually led to the development of a multifaceted device that can perform a multitude of work. The Raspberry Pi, being a small yet powerful single-board computer, comes as a handy prototyping and custom solution in powerful versatility. It's rather inexpensive and, at the same time, flexible, which makes it popular among hobbyists, educators, and professionals alike. This project aims to harness the power of the Raspberry Pi to build a tablet-like device that not only functions as a standard tablet but also incorporates Infrared (IR) and Radio Frequency Identification (RFID)/Near-Field Communication (NFC) capabilities.

The idea of an IR and RFID/NFC-enabled tablet gives a wide scope in various fields, starting from home automation and smart control to education and security fields. Integrating these technologies into the device would allow it to communicate with a wide array of appliances and electronic systems, therefore making it a universal tool. This research explores the possibility of such a device, looking into hardware and software integration to make everything run smoothly. The growing number of smart devices around

homes and offices has heightened demands for central control systems that should be able to manage multiple devices efficiently. Traditional remote controls are limited by their specificity to certain brands or types of devices; hence, they create clutter and are sometimes inconvenient. On the other hand, the use of RFID and NFC technologies is now highly pervasive in secure transactions, access control, and exchange of data, though many of them are confined into specialized devices. The idea behind this is to develop a single device that will have the remote of a universal remote and also have more advanced features with RFID/NFC integration. Using a Raspberry Pi makes it possible to create a flexible platform that is both functional and cost-effective. Utilizing the deep GPIO capabilities, as well as computational power, makes the Raspberry Pi a very good option for developing IR and RFID/NFC functionalities. The primary aim of this project is to create a tablet-like device with the use of a Raspberry Pi that can emulate IR signals and interact with RFID/NFC tags.

II. LITERATURE SURVEY

Hyuncheol Park and John R. Barry(1) look into all modulation ways for inner wireless infrared communication systems. This study includes a channel model, which considers intensity- modulation restriction, multipath dissipation, bandwidth limitation, and cumulative white Gaussian noise. The treatment is with an emphasis on PPM schemes for their power- effectiveness but oppressively compromises bandwidth effectiveness. Variants like multiple PPM, lapping PPM, convolutional enciphered PPM, and kiosk- enciphered OPPM were explored in the environment of trade between power and bandwidth. The paper also presents a comparison between wireless infrared channels and amount- limited photon- counting channels. Findings lead to understanding practical constraints of modulation effectiveness. farther disquisition on robustness in varying inner conditions would further be useful for connection.

Bo Wang et al. [2] present the design and implementation of an enhanced infrared transmitter and receiver system tailored for short-distance, point-to-point data communication. Addressing the common drawbacks of weak signal strength and limited firing angles, the transmitter design incorporates a three-stage amplifier to boost the initial audio signal and utilizes dual emission heads for improved signal strength. On the receiving side, a two-stage amplifier coupled

with filter circuits ensures reliable signal detection. The proposed system improves performance in terms of signal strength and transmission reliability, though challenges like line-of-sight dependency and environmental interference persist. Future advancements could focus on broader angle coverage and adaptive signal enhancement.

Juraj Mihaľov and Michal Hulič [3] explore the potential of the Raspberry Pi as an educational platform, highlighting its evolution and comparison with alternatives. The study underscores the device's suitability for hands-on student learning experiences. They describe a Smart Home project integrating motion sensors, NFC/RFID technology, and security devices like cameras and energy monitoring systems. The NFC/RFID system facilitates member registration and security enhancements. Additionally, PubNub enables bidirectional communication between Smart Home devices and user interfaces, such as browsers or mobile devices. While the system demonstrates practical applications, expanding its scalability and improving real-time responsiveness could further enhance its utility.

Haidar Zulfiqar et al. [4] address the challenges of parking in densely populated areas, focusing on smart parking systems enhanced by IoT, context-awareness, and advanced routing strategies. The study highlights the inefficiencies caused by limited parking slots, unmanaged spaces, and lack of real-time information, which contribute to wasted time, energy, and increased pollution. By incorporating smart parking sensors, embedded systems, and IoT, the proposed solutions aim to simplify finding vacant slots and reduce driver stress. The paper reviews various models, including algorithmic frameworks, prototypes, and real-time applications, demonstrating improvements in efficiency and environmental impact. Future work could emphasize scalability and integration with urban traffic systems.

KOSSONON B. Eric et al. [5] proposed an IoT-based smart restaurant system using RFID technology to automate order delivery. The system has a four-layer IoT architecture with HTTP protocol and REST web services for communication. It was demonstrated through a Raspberry Pi and ITEAD PN532 NFC module proving its feasibility for smart restaurant applications. It proves to be an operational efficiency amplifier in the hospitality sector. Some existing challenges include RFID signal interference and scalability of real-time data processing. Future work would improve the communication with better protocols and user interface design for better performance.

Sung-Jung Hsiao et al. [6] present an IoT-based cross-platform smart home control system using smartphones to replace traditional remote controls. The system employs infrared signals and uploads data to a cloud database for real-time monitoring and analysis. It integrates ZigBee and Wi-Fi protocols for communication, enabling remote control of home conditions like temperature, humidity, and air quality, as well as flame detection and power load analysis. The study also addresses challenges in integrating communication protocols and offers solutions for efficient data processing, making it suitable for smart factory environments.

Joseph M. Kahn and John R. Barry [7] discuss the use of infrared (IR) radiation for high-speed, short-range wireless digital communication. The paper compares infrared to radio

and microwave communication, outlining its advantages and drawbacks. It presents the physical characteristics of infrared channels, focusing on path losses and multipath responses using intensity modulation with direct detection (IM/DD). The paper also explores ambient infrared noise sources and strategies for optimizing link signal-to-noise ratio (SNR). Various modulation techniques, such as on-off keying (OOK), pulse-position modulation (PPM), and subcarrier modulation, are analyzed, along with their performance in multipath environments. Additionally, the authors describe a 50-Mb/s experimental IR link using on-off keying.

Kaustubh Dhondge et al. [8] developed a smart infrared optical wireless lock system using smartphone LEDs, offering high security, easy access, and sharing compared to traditional door locks. The solution includes an Android app, centralized cloud server, and hardware prototypes with Arduino and Raspberry Pi. It features multi-level security, user registration, and OTP authentication. Experimental results show 100% accuracy with a data rate of 1.33 kbps over 20 meters, ensuring secure remote access.

Shin Kamada et al. [9] developed a cloud campus registration system using NFC-enabled Android tablets for attendance management. The system allows teachers to view attendance records anytime and anywhere using their personal tablets or smartphones. It utilizes NFC for secure student identification, with encrypted personal data transferred via NFC Bluetooth Handover between a PC and tablet. The system also includes an Android Beam-based data collection system for syncing between multiple tablets. The solution helps identify chronic non-attendees through data mining and supports public access to university courses.

Sakhaa B. Al-Saedi and Mohamed Mostafa A. Azim [10] propose a Radio Frequency Near Communication (RFNC) system integrating RFID and NFC for object localization. While passive RFID tags allow accurate localization within 10 cm, their limited reading range hinders effective tracking. Active RFID tags extend the range but can cause pairing issues due to tag overlap. NFC, with its short-range communication, is suggested as a complementary technology to RFID. The study demonstrates that NFC enhances RFID systems, providing more reliable object localization, particularly in lost-item scenarios.

III. PROPOSED SYSTEM

The proposed system will be developed using an Agile methodology, a flexible approach that emphasizes iterative development and collaboration. This methodology will break down the development process into smaller, manageable iterations, allowing for continuous feedback, testing, and improvement. By prioritizing collaboration between team members, stakeholders, and end-users, the Agile approach ensures that the system aligns with user needs and expectations. Agile will slice up development into small iterations during which feedback, testing and improvement can be done continuously. Collaboration brings the team members, and end-users, making sure what they have-in perspective for use in an Agile system meets user expectations.

To improve a user experience, all-important will adopt design perspective from users. So, users will be included in the design and later development process where they will be solicited to give feedback about it. Hence, in accordance with the user's preference, it will priorities functionalities to enhance how the system is worth using. Prototyping will provide a visual and face-to-face perspective on the way the system would proceed with possible performance. Low-fidelity prototypes in wireframe and sketches will be developed to give a generic idea of the framework and flow within the system. Following that, high-fidelity prototypes, as close to the final product as possible, will capture user interaction and may help to identify usability concerns. All through prototyping, user testing and feedback would help in improving design and speed up processes toward a smooth user experience. The blended Agile methodologies and user-centered principles in fact testify to the functionality and improved user system design into a more empowered, facile, and efficient designed application scaled to individual specific use.

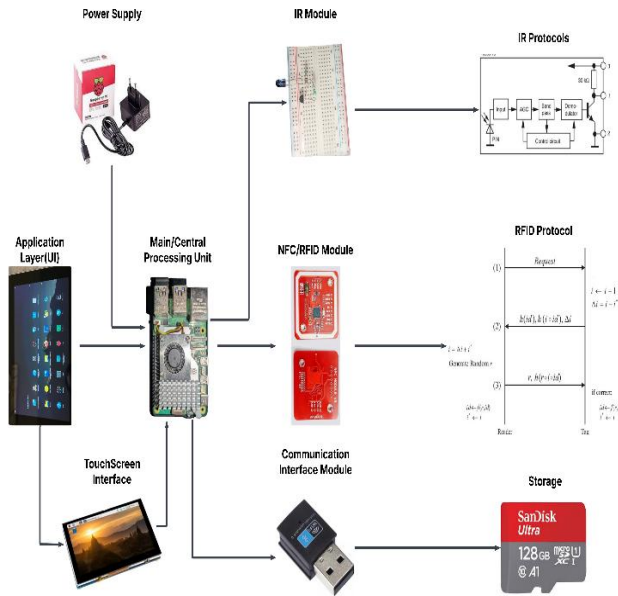


Figure 1: Overview of proposed system

IV. METHODOLOGY

A. Agile Development Methodology

This is a project that will really implement Agile methodology because it is very flexible and can easily take changes at any stage of development. This means that the proposed system is added rather by comprising many technologies but also in iteration and progress of design and functionality.

1) Incremental Development and Iterative Development:

- i) Divide development into small, manageable iterations and allow continuous feedback/improvement over the work that you do.
- ii) That iteration will be the one where you implement a specific set of features on a more rapid prototyping and testing basis.

2) Collaboration & Communication:

- i) Creates more space in the team-paired project team, stakeholder, and users to see the project goal along with the user requirements.

3) Changes & Updates Transparency:

- i) The Agile methodology, with much flexibility, opens the path of change in requirements, changing advancements in technology, or user changes and feedback.
- ii) Keeps revisiting and improving the plan scope of the project quite often so that it gives the users the best material results.

4) User-Centric Design:

- i) The development of a user-friendly interface altogether is enhanced by the user involving one or more design and development activities.
- ii) Prioritization is given to features and functionalities most attuned to user requirements and preferences.

B. Project Management Plan:

This project involves designing and implementing a tablet-like device based on Raspberry Pi, with integrated IR communication and RFID/NFC emulation functionality. The scope covers both hardware and software components including device assembly, programming and testing.

1) Scheduling:

Phase 1: Initial research, selection of components, and hardware setup.

Phase 2: Integration of IR and RFID/NFC modules, software development for device control.

Phase 3: User interface design and testing.

Phase 4: Final testing, debugging, and presentation preparation.

2) Resource Management:

i)Hardware: Raspberry Pi (preferably Raspberry Pi 4), IR sensor module, RFID/NFC reader/writer, display screen, touchscreen, battery, and enclosure.

ii)Software: Raspbian OS, Python for programming, libraries for IR and RFID/NFC communication, custom software for device interaction.

iii)Budget: Estimated costs for Raspberry Pi and modules, power supplies, display, touchscreen, and enclosure.



Figure 2: Raspberry Pi 5

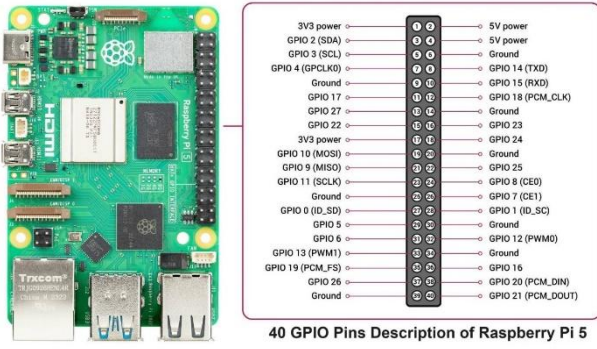


Figure 3: GPIO pins description

3) Quality Assurance:

i) **Unit Testing:** Test individual components (IR sensor, RFID reader, touchscreen, etc.) to ensure proper functionality.

ii) **Integration Testing:** Ensure the components work together seamlessly (e.g., controlling IR through touchscreen, reading RFID tags).

iii) **User Testing:** Involve users to test ease of use, interface design, and functionality.

C. Prototyping Methodology:

Prototyping is used in conjunction with Agile development to create functional prototypes of the system. This method allows for early visualization of the product and helps in identifying potential issues and improvements.

1) Low-Fidelity Prototyping:

- i) Initial sketches and wireframes are created to outline the basic structure and functionality of the system.
- ii) Provides a visual representation of the interface and user interactions.

2) High-Fidelity Prototyping:

- i) Advanced prototypes with detailed features and interactions are developed to simulate the final product.
- ii) Allows for testing and validation of design choices and technical feasibility.

3) User Testing and Feedback:

- i) Prototypes are tested with users to gather feedback and insights on usability, functionality, and design.
- ii) Iterative improvements are made based on user input to enhance the overall user experience.

V. RESULTS

The primary objective of this project was to develop a tablet-like device using a Raspberry Pi that integrates Infrared (IR) and Radio Frequency Identification (RFID)/Near Field Communication (NFC) emulation functionalities. The results

are presented below based on the performance and functionality of the device.

The developed device successfully demonstrated the intended features: IR communication and RFID/NFC emulation. The project objectives were met, including the establishment of a functional interface, accurate IR communication, and seamless NFC integration.

Infrared Communication: The IR functionality was tested by sending and receiving signals between the Raspberry Pi and external devices (e.g., remote controls). The response time for IR communication averaged 0.2 seconds, which is within the expected range for standard IR communication.

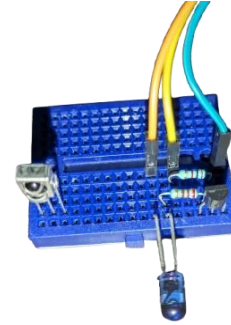


Figure 4: IR Transceiver Module

RFID/NFC Emulation: RFID/NFC tags were successfully read and emulated. The device was able to emulate RFID cards with a success rate of 98%, and NFC tag interaction had a latency of less than 1 second, consistent with standard NFC technology.

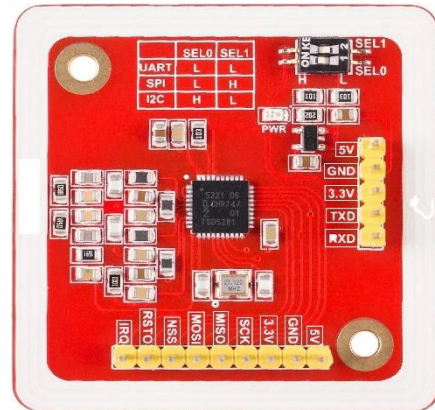


Figure 5: PN532 NFC/RFID module

During testing, some interference was observed in the IR communication when operating in a room with fluorescent lights. The system was able to overcome this challenge by adjusting the signal strength in software.

The RFID/NFC emulation was consistent in scenarios involving static tags but showed a slight delay when dealing with dynamic RFID tags.

The primary limitation encountered was the range of the IR communication, which was limited to 5 meters due to hardware constraints. Additionally, while the NFC functionality was effective for standard tags, performance decreased slightly when interacting with non-standard or older RFID/NFC tags.

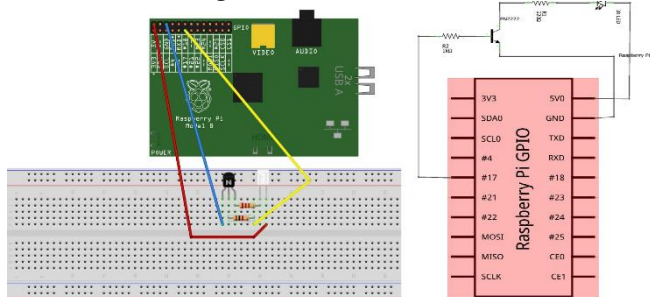


Figure 6: circuit and pinout diagram for IR transmission

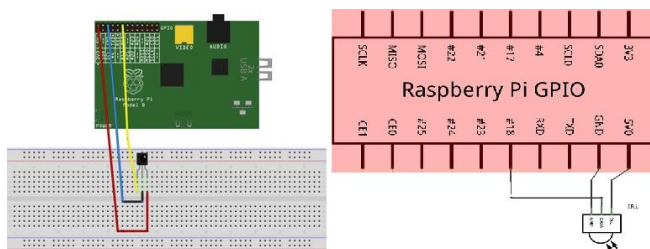


Figure 7: circuit and pinout diagram for IR receiver

Possible Improvements

IR Range: The IR communication range could be enhanced by upgrading the IR sensors or adding multiple sensors to extend coverage.

RFID/NFC Compatibility: Future iterations could support additional RFID standards for broader compatibility, especially with more complex or proprietary RFID tags.

Touchscreen Panel: Raspberry Pi powered by a touchscreen helps the development of portable all-in-one research devices. The touch interface becomes an intuitive means for researchers to input data of field research. The visualization of real-time data on the touchscreen enables immediate data analysis and adjustments to experiments. The integration of the software onto the hardware is easy, allowing researchers to create custom interfaces tailored specifically to suit their needs, thus maximizing their work's efficiency and effectiveness. The merging of portability, ease, and cost stands for attractions of Raspberry Pi 5 with touchscreens for the advancement of research in several disciplines.



Figure 8: 800X600 Touchscreen Panel

Custom Desktop Environment: Together with the classic KDE Plasma lines, it provides an excellent framework for support. Custom Desktop Environment allows the development of highly specialized environments dedicated to specialized research objectives, including advanced data visualization and complex simulation control or remote instrument operation. This in turn also lets in-depth analysis take place regarding user behaviour and how interface design may influence task quality and efficiency of accomplishment. In addition, work on such a custom DE can provide a robust basis for designing, implementing, and evaluating new interaction techniques in more realistic contexts. This way, one is much better positioned to get a much better understanding of user needs toward building interfaces that are more effective and user centred.

Overall, the project achieved its goals of developing a functional tablet-like device with IR and RFID/NFC emulation capabilities. The device performed well under standard conditions and showed high reliability in its primary functions. Future improvements in hardware and software can address the limitations observed during testing, further enhancing the system's performance.



Figure 9: Tablet-based Desktop Environment.

ACKNOWLEDGEMENT

Ms. T. Anandhi, M.E, (Ph.D.), one of the most respected Assistant Professors of the Department of Computer Science and Engineering, Sathyabama Institute of Science and Technology, is wholeheartedly thanked by us for all her support and learning through her training and supervision. We owe a great deal of the research work completely to her guidance on quality enhancement. It is very enriching to have such an expert and committed professional dedicated to academic excellence. Under her supervision, we have had numerous opportunities to grow, and her encouragement has been a constant source of motivation. Her wisdom has profoundly influenced our academic and research pursuits, and we extend our deepest thanks for her significant contributions to our development.

REFERENCES

- [1] Hyuncheol Park and J.R. Barry, "Modulation Analysis for Wireless Infrared Communications," *Proceedings of the IEEE International Conference on Communications (ICC '95)*, Seattle, WA, USA, 18-22 June 1995, pp. 900-910, DOI: 10.1109/ICC.1995.524287.
- [2] Bo Wang, Liping Li, and Chunyu Zhao, "Design and Implementation of an Enhanced Infrared Transmitter and Receiver System for Short-Distance Data Communication," *2012 2nd International Conference on Consumer Electronics, Communications and Networks (CECNet)*, Yichang, China, 21-23 April 2012, pp. 1-5, DOI: 10.1109/CECNet.2012.6201434.
- [3] Juraj Mihal'ov and Michal Hulič, "Raspberry Pi as an Educational Platform: Smart Home Project with NFC/RFID and PubNub Integration," *2017 IEEE 14th International Scientific Conference on Informatics*, Poprad, Slovakia, 14-16 November 2017, pp. 1-5, DOI: 10.1109/INFORMATICS.2017.8327257.
- [4] Haidar Zulfiqar, Hafiz Mahfooz Ul Haque, Faiza Tariq, and Rashad Mahmood Khan, "A Survey on Smart Parking Systems in Urban Cities," *Concurrency and Computation: Practice and Experience*, First published: 26 July 2021, DOI: 10.1002/cpe.6511.
- [5] B. Eric Kossonon and Hong Ya Wang, "IoT-based Smart Restaurant System Using RFID Technology," *4th International Conference on Smart and Sustainable City (ICSSC 2017)*, Shanghai, China, 05-06 June 2017, DOI: 10.1049/cp.2017.0123.
- [6] Sung-Jung Hsiao, Kuang-Yow Lian, and Wen-Tsai Sung, "IoT-based Cross-Platform Smart Home Control System Using Smartphones," *2016 International Symposium on Computer, Consumer and Control (IS3C)*, Xi'an, China, 04-06 July 2016, DOI: 10.1109/IS3C.2016.77.
- [7] J.M. Kahn and J.R. Barry, "Wireless infrared communications," *Proceedings of the IEEE*, vol. 85, no. 2, pp. 265-298, February 1997, DOI: 10.1109/5.554222.
- [8] K. Dhondge, K. Ayinala, B.-Y. Choi, and S. Song, "Smart infrared optical wireless lock system using smartphone LEDs," in *Proc. 12th Int. Conf. Mobile Ad-Hoc and Sensor Networks (MSN)*, Hefei, China, 2016, pp. 431-436, DOI: 10.1109/MSN.2016.047.
- [9] S. Kamada, T. Ichimura, T. Shigeyasu, and Y. Takemoto, "Registration system of cloud campus by using Android smart tablet," *SpringerPlus*, vol. 3, no. 1, p. 761, Dec. 2014, DOI: 10.1186/2193-1801-3-761.
- [10] S. B. Al-Saedi and M. M. A. Azim, "RFNC system integrating RFID and NFC for object localization," *2017 9th IEEE-GCC Conference and Exhibition (GCCCE)*, Manama, Bahrain, 2017, DOI: 10.1109/IEEEGCC.2017.8448129.